

Introduction to Calculations of Heat loss on piping

Przemyslaw Wiecek

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1 Introduction

At some point during one of the project I was a part of, it was necessary to check if the chosen thickness of insulation for long CIP water loops was enough, to keep the temperature control and over certain threshold. It turned out that one of calculations were done in a rush, resulting in wrong decision of making insulation thicker. Just in time, one of senior engineers stepped in and stopped the company from making somewhat pricy mistake.

This senior engineer, later send us the calculaitons to check it, write down things that were wrong and report back to him, to have a conversation about our findings. This way we could learn to spot errors. Because I like calculations and love learning about things inherently related to mathematics it was a match made in heaven. At this point in time I have not started my introductory book to mechanics of heat transfer by great J. Cengel, so it wasn't something I could jump right into. Thus, I took my time, and in around 2 weeks felt somewhat comfortable, at least with concepts (side note: I was especially taken by transient analysis and eigendecomposition of differential equations in plug flow problems, amazing, it might be something to write about) in steady state analysis to begin deriving equations and testing my knowledge.

The original calculations were made using iterative method, which is fine if the iterations are "fairly" small or the change in the underlying function is relatively insignificant or, even better, constant.

2 Notation

The functions we will be dealing with depends inherently on a lot of variables, some of them are constant no matter the distance the fluid traveled, others are not. In general the notation for each function F can be presented as:

$$F(x, \mathbf{v}) = F(x, \mathbf{u}(x), \mathbf{v}, \mathbf{s})$$

With $\mathbf{v} \in \mathbf{R}^N, \mathbf{u} \in \mathbf{R}^M, \mathbf{s} \in \mathbf{R}^J$. Where N is the number of variables that are independent of x , M is the number of variables that depend explicitly on x and J is the number of variables that are independent of x and constant.

Let us use combined thermal conductivity K as an example. For outlined problem K is a function of form:

$$K(x, h_{comb}) = K(D_o(x), D_i(x), y(x), h_{comb}, h_{f \rightarrow w}, k_{pipe}, k_{ins})$$

If this dependence is not outlined in any form then, we state that K is constant over total length. Since, when it comes to piping, diameter does not change constantly with length, and we are dealing with segments of different length of constant diameter. We will be making a lot of reference to those discrete points, either to segment lengths or to the cumulative length up to given segment. The notation used is:

ith segment length: x_i^s

$$\text{Cumulative length up to } nth \text{ segment: } x_n = \sum_{i=0}^n x_i^s$$

There are places where an equation can get somewhat long, and so it is helpful to introduce another notation that will help keep those equations tidy. Again, taking K as an example:

$$K_i = K(x_i, h_{comb})$$

With that out of the way, let us proceed to the fun part.

3 Derivation

3.1 Basics of steady-state heat transfer

The assumptions we must consider to make calculations possible (and easy) are:

1. Steady-state operation (both mass and energy).
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid in question is incompressible.
5. The specific heat capacity is constant.
6. No axial heat transfer.
7. Piping is horizontal and enclosed by air only at ambient pressure.

With above in mind, one can utilize "Thermal Resistance Network" methodology and write that the average heat loss rate through a pipe on a segment of length L and distance x from the inlet is:

$$\dot{Q}_{avg}(x, h_{comb}) = \frac{(T(x, h_{comb}) - T_{2\infty})_{avg}}{R(x, h_{comb}, L)} = 2\pi L K(x, h_{comb}) (T(x, h_{comb}) - T_{2\infty})_{avg} \quad (1)$$

The *avg* notation means that we should take average of the temperature difference on piping segment of length L . This is mathematically explained in equation (4). With the same assumptions we can write that energy balance at any point must satisfy:

$$\dot{m}CdT + \dot{Q} = 0 \quad (2)$$

where:

- $T(x)$ - temperature of liquid at distance x , (F°)
- $T_{2\infty}$ - ambient temperature, (F°)
- $R(L)$ - thermal resistance of piping of length L , $\left(\frac{F^\circ}{BTU}\right)$

- K - combined thermal conductivity, $\left(\frac{BTU}{ft \cdot h \cdot F^\circ}\right)$
- $\dot{m}$ - mass flow rate, $\left(\frac{lbm}{h}\right)$
- C - specific heat capacity, $\left(\frac{BTU}{lb \cdot F^\circ}\right)$

and we define K as:

$$\frac{1}{K} = 2\pi LR(L) = \frac{24}{h_{f \rightarrow w} D_i} + \frac{1}{k_{pipe}} \ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} \ln\left(\frac{D_o + 2y}{D_o}\right) + \frac{24}{h_{comb}(D_o + 2y)} \quad (3)$$

where:

- y - insulation thickness, *inch*
- D_i - inner diameter, *inch*
- D_o - outer diameter, *inch*
- $h_{f \rightarrow w}$ - heat transfer coefficient from fluid to piping wall, $\left(\frac{BTU}{ft^2 \cdot h \cdot F^\circ}\right)$
- h_{comb} - combined heat transfer coefficient from surface of a piping to air, $\left(\frac{BTU}{ft^2 \cdot h \cdot F^\circ}\right)$
- k_{pipe} - thermal conductivity of a piping, $\left(\frac{BTU}{ft \cdot h \cdot F^\circ}\right)$
- k_{ins} - thermal conductivity of a insulation, $\left(\frac{BTU}{ft \cdot h \cdot F^\circ}\right)$

3.2 General solution

For simplicity let us drop h_{comb} from functions in the derivation. So, when we write $F(x)$ we really mean $F(x, h_{comb})$. We begin our analysis with finite distance change. Using graph for this would be the simplest approach.

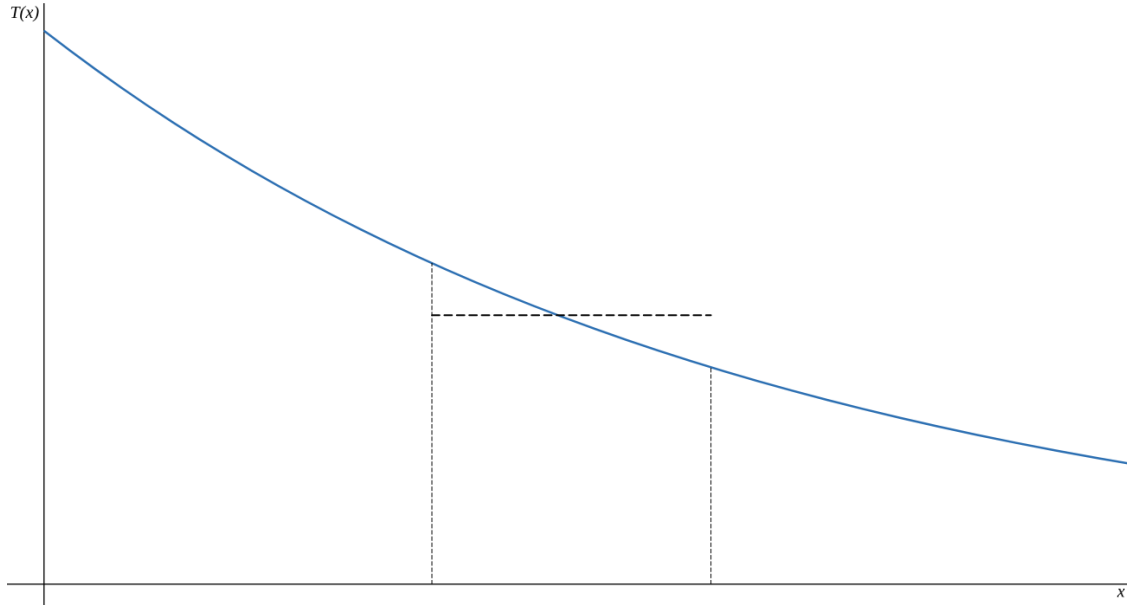


Figure 1: $T(x)$ Function of $T(x)$

In the figure above we consider a change in temperature over a finite change in length on interval $[x_i, x_{i+1}]$, where temperature drops from $T(x_i)$ to $T(x_{i+1})$. The heat loss over piping segment $x_{i+1} - x_i = \Delta x_i$ is not constant, but drops with temperature. Because of that we consider some average temperature difference to find the average heat loss. The temperature we will use to find the difference is shown on the figure above as a horizontal line between x_i and x_{i+1} . With $w_i \in [0, 1]$ we can write that the average heat loss is:

$$\dot{Q}_{avg}(x_i) = 2\pi(x_{i+1} - x_i)K_i \cdot (T(x_i) - T_{2\infty})_{avg} = 2\pi\Delta x_i K_i \cdot (T(x_i + w_i\Delta x_i) - T_{2\infty}) \quad (4)$$

And energy balance:

$$-\dot{Q}_{avg}(x_i) = \dot{m}C \cdot (T(x_{i+1}) - T(x_i)) = \dot{m}C\Delta T_i \quad (5)$$

We connect eq (4) and (5), rearrange it and write $x_i^* = x_i + w_i\Delta x_i$:

$$\frac{1}{T(x_i^*) - T_{2\infty}} \frac{\Delta T_i}{\Delta x_i} = -\frac{2\pi K_i}{\dot{m}C} = -\frac{K_i}{\lambda} \quad (6)$$

Remembering that by $T(x)$ we mean $T(x, h)$ we can finally write:

$$\lim_{\max(\Delta x_i) \rightarrow 0} \frac{1}{T(x_i^*, h) - T_{2\infty}} \frac{\Delta T_i}{\Delta x_i} = \frac{1}{T(x, h) - T_{2\infty}} \frac{\partial T(x, h)}{\partial x} = -\frac{K(x, h)}{\lambda} \quad (7)$$

And the general solution after some cleaning can be written as:

$$\frac{\partial \ln(T(x, h) - T_{2\infty})}{\partial x} = -\frac{K(x, h)}{\lambda} \quad (8)$$

Integrating LHS from T_0 to $T(x, h)$ and RHS from 0 to x yields:

$$\ln\left(\frac{T(x, h) - T_{2\infty}}{T_0 - T_{2\infty}}\right) = -\frac{1}{\lambda} \int_0^x K(l, h) dl \quad (9)$$

To solve it we need to know how $K(x, h)$ changes with length. We would also need to show that h is not a function of x . If we take a look at K it's obvious we can vary h independently of x , but we assumed steady-state conditions, and if this forces h to be dependent on x then it would render solving this integral hard or even impossible.

I was not able to prove it mathematically but in every tested scenario I tried h is independent of x . All my calculation and methodology used is described in chapter (X). For the time being we assume that h is independent and proceed with solving eq (9) for each case.

ANOTHER TAKE: I was not able to prove it mathematically but I based my argument on statistical interpretation of our assumptions as well as results of used heat transfer coefficients correlations. All my calculation and methodology used is described in chapter (X). For the time being we assume that h is independent and proceed with solving eq (9) for each case.

3.3 Solutions $h_{comb} \rightarrow \infty$

We start our approach to solving eq (9) by letting $h_{comb} \rightarrow \infty$. This eliminates h_{comb} from K providing us with simple solutions. The main consequence is that surface temperature of insulation is driven to ambient temperature around piping, which maximizes temperature gradient maximizing heat transfer in turn. This method provides the most conservative solution and yields the biggest temperature drop. This method was used in my excel template for the company I'm working for as "conservative" approach, providing user with quick and simple answer to the question "is currently selected insulation thickness enough for the worst-case scenario?".

3.3.1 Constant diameter

We start with a long pipe of length L_t of constant inner and outer diameter D_i, D_o , respectively as well as constant insulation thickness y . Those assumptions allows us to say that K does not depend on x and so we can take it out of the integral. The solution is, then:

$$\ln \left(\frac{T(x) - T_{2\infty}}{T_0 - T_{2\infty}} \right) = -\frac{K}{\lambda} x \quad (10)$$

for $x \in [0, L_t]$. Short and sweet. Not much to say, and so we move to the next scenario.

3.3.2 Segments with different diameter and insulation thickness

Here we let D_i, D_o and y vary over specified segments with length. We have to figure out the form of $K(x)$. It is of course just a piecewise defined function:

$$K(x) = \begin{cases} K(x_0) & \text{if } 0 < x \leq x_0 \\ \cdot \\ \cdot \\ \cdot \\ K(x_n) & \text{if } x_{n-1} < x \leq x_n \end{cases} \quad (11)$$

This form provides a simple way of solving the integral from eq (9). Initially, unfortunately, I wrote my whole python code with heaviside function, because I feared that above approach was too convoluted, little did I know, I spend few months writting and working with only-functions code, in the end producing piece of software hard to build upon. Any new idea was met with some rewrites and then long sessions of figuring out why, again, heaviside funciton part does not work. The solution was hard to grasp and very slow. Running few thousands calculations for some analysis required waiting tens of minutes. In the end, one day, I figured why not just go with what's above, and I just rewrote my whole mainframe. The code was clean(er) easier to understand, and so much faster, oh so much... tens of thousands of calculations in less than 10 seconds. Nice.

With this simple form of eq (11) we can almost solve eq (9). Since, we are working on different segments now, we must solve them separately. Thus we must go back to eq (8) and integrate it for every segment, which yields in general:

$$\ln \left(\frac{T(x) - T_{2\infty}}{T(x_i) - T_{2\infty}} \right) = -\frac{1}{\lambda} \int_{x_i}^x K_i(l) dl = -\frac{K_i}{\lambda} (x - x_i) \quad (12)$$

For $x \in [x_i, x_{i+1}]$. Here it is somewhat simple to see that given logarithm properties we can obtain a whole piping equation from:

$$\ln \left(\frac{\Delta T(x)}{\Delta T(x_0)} \right) = \ln \left(\frac{\Delta T(x)}{\Delta T(x_{k-1})} \right) + \sum_{i=0}^{k-1} \ln \left(\frac{\Delta T(x_{i+1})}{\Delta T(x_i)} \right) = -\frac{1}{\lambda} \left(K_{k-1}x + \sum_{i=0}^{k-1} K_{i-1}x_i \right) \quad (13)$$

I didn't use it at any point when rewriting the code as class based approach didn't require it at any point. So, it is more of an interesting solution.

This is the "conservative approach" method, yielding the biggest temperature drop accross the whole pipe. It is worth mentioning that for thick enough insulation, this method gives preety accurate results, but "thick enough insulation" is enought of an ambiguous statement to not mean anything in particular. When thickness of insulation is close to 0, this method brings fulid temperature to ambient temperature in matters of meters, and would be highly disregarded for those type of applications where accuracy is needed.

3.4 Exact solutions

The heat rate in this example is a funciton of position, given cylindrical coordinates we can write that:

$$\dot{Q} = \dot{Q}(r, \theta, z) \quad (14)$$

We have assumed steady-state heat transfer through piping wall, thus we can drop r relation. We also assumed that heat transfer is uniform through the piping, so θ can be dropped as well. Then, we are left with only z direction, which we were referring to as x . Steady-state heat transfer tells us also that heat transfer in the cross section is the same in magnitude no matter at which r it is measured. Mathematically:

$$\dot{Q}_r(x) - \dot{Q}_{r+\Delta r}(x) = 0 \quad (15)$$

This is exactly the condition we can use to solve for the fluid temperature. All we need is to choose which heat transfer to use. More on that later. First off, it is much easier to utilize eq (1) going further, as it represents heat rate in more natural local properties. One change is that instead of using heat transfer, we will be going with heat rate per reference length L , which simplifies calculations.